Taiwan Cancer Registry

Temporal Trends in Cancer Incidence 2003–2020

Overview

Using first-primary cancer records for 78,619 patients registered 2003–2020, we computed annual site-specific fractions and tested monotone trend (Spearman ρ). p-values are autocorrelation-corrected (Chelton 1983; lag-1 AR removed from residuals; effective n computed as n*(1-φ)/(1+φ)).

Hypothesis Results

H1 C22 liver HCC declining → CONFIRMED
    ρ=−0.983, p_corr=<0.001 (φ=0.43, n_eff=7)

H2 UADT betel sites declining post-2006 → SPLIT
    C13 hypopharynx: ρ=−0.773, p_corr=<0.001 (falling)
    C12 pyriform: ρ=+0.562, p_corr=0.0042 (rising)

H3 Hormonal sites rising (metabolic syndrome) → PARTIAL
    C54 endometrial: ρ=+0.925, p_corr=0.0085 (confirmed)
    C50 breast: ρ=-0.238, p=0.3408 (ns, φ=0.77)

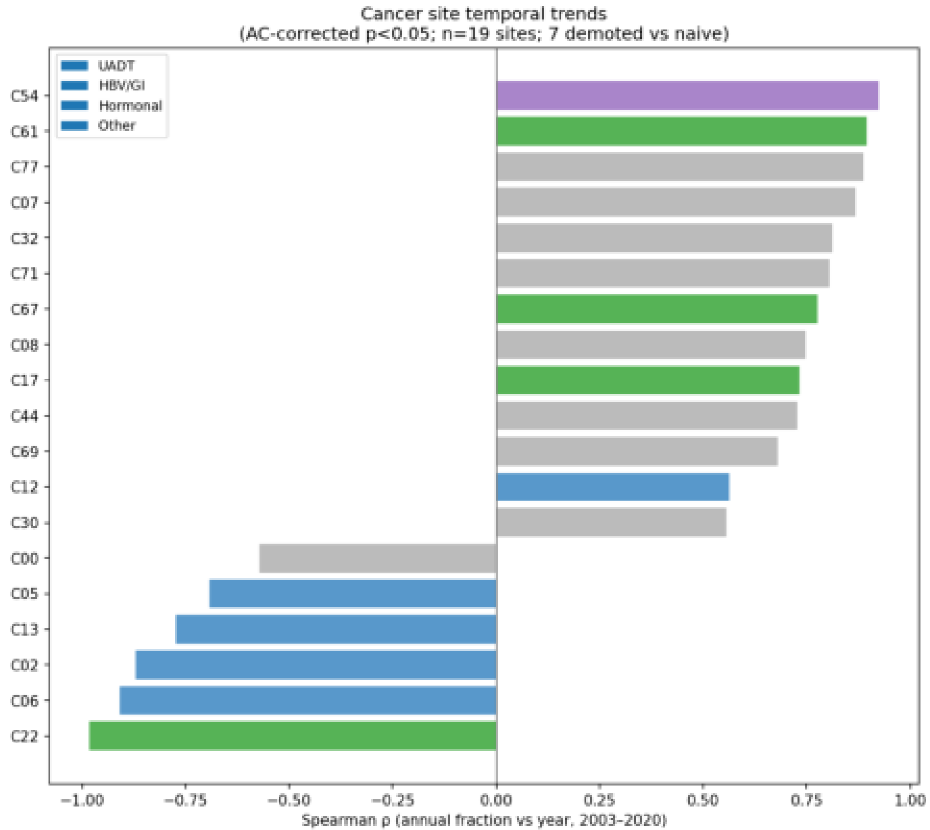
Bonus Findings

    C53 cervix: ρ=−0.946 p_corr=0.2095
    C61 prostate: ρ=+0.895 p_corr=0.0101

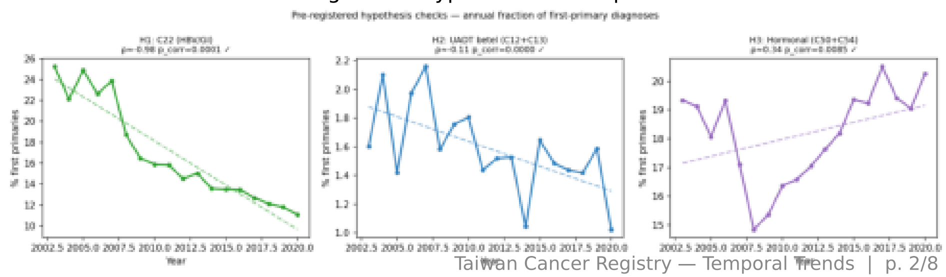
    AC-corrected: 13 rising, 6 falling (p_corr<0.05)
    Naive p<0.05 demoted by AC correction: 7 sites

Site-level temporal trends — overview

All sites: Spearman ρ (annual fraction vs year)

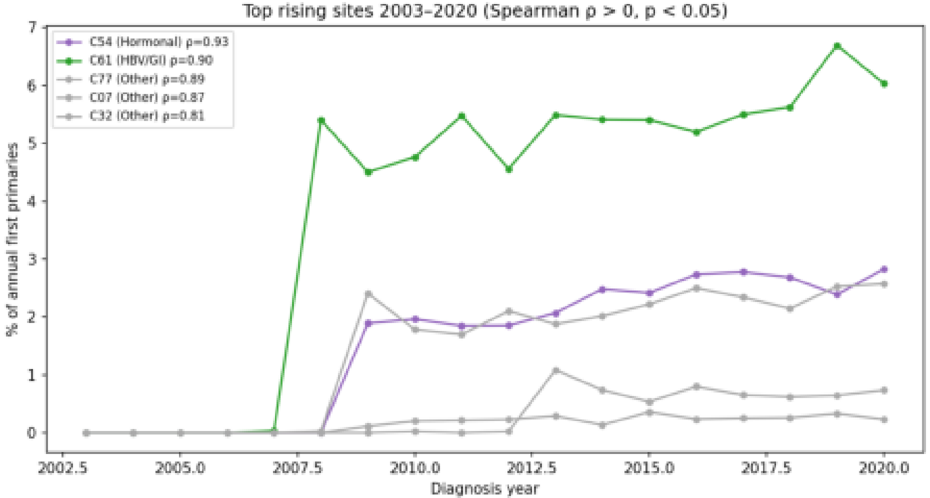


Pre-registered hypothesis check panels

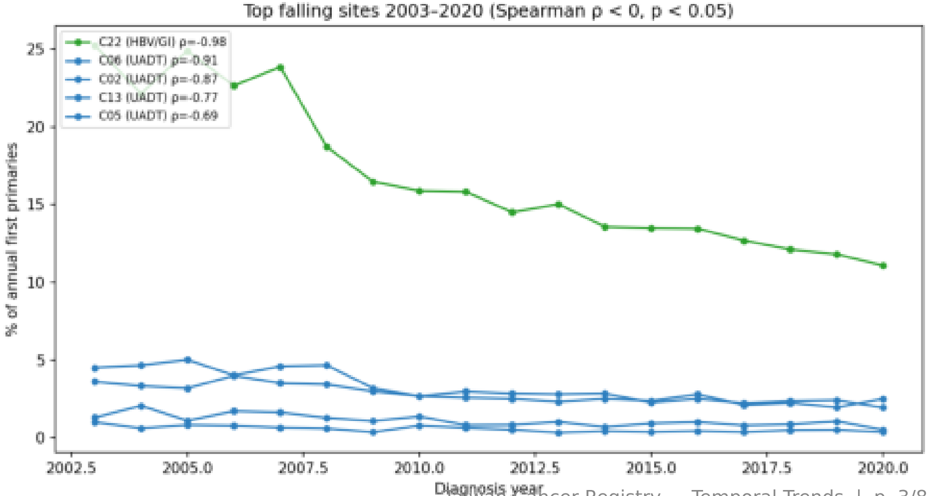


Rising and falling sites — annual fraction of first primaries

Top rising sites (p<0.05)

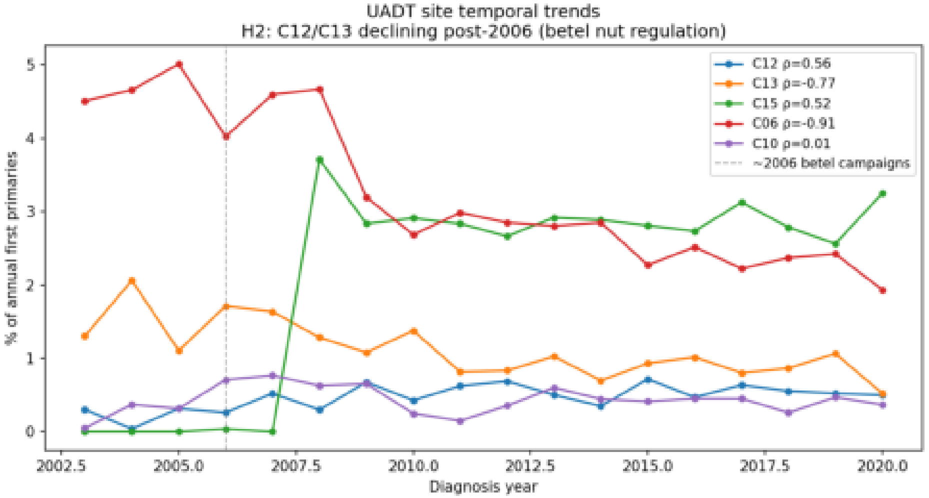


Top falling sites (p<0.05)

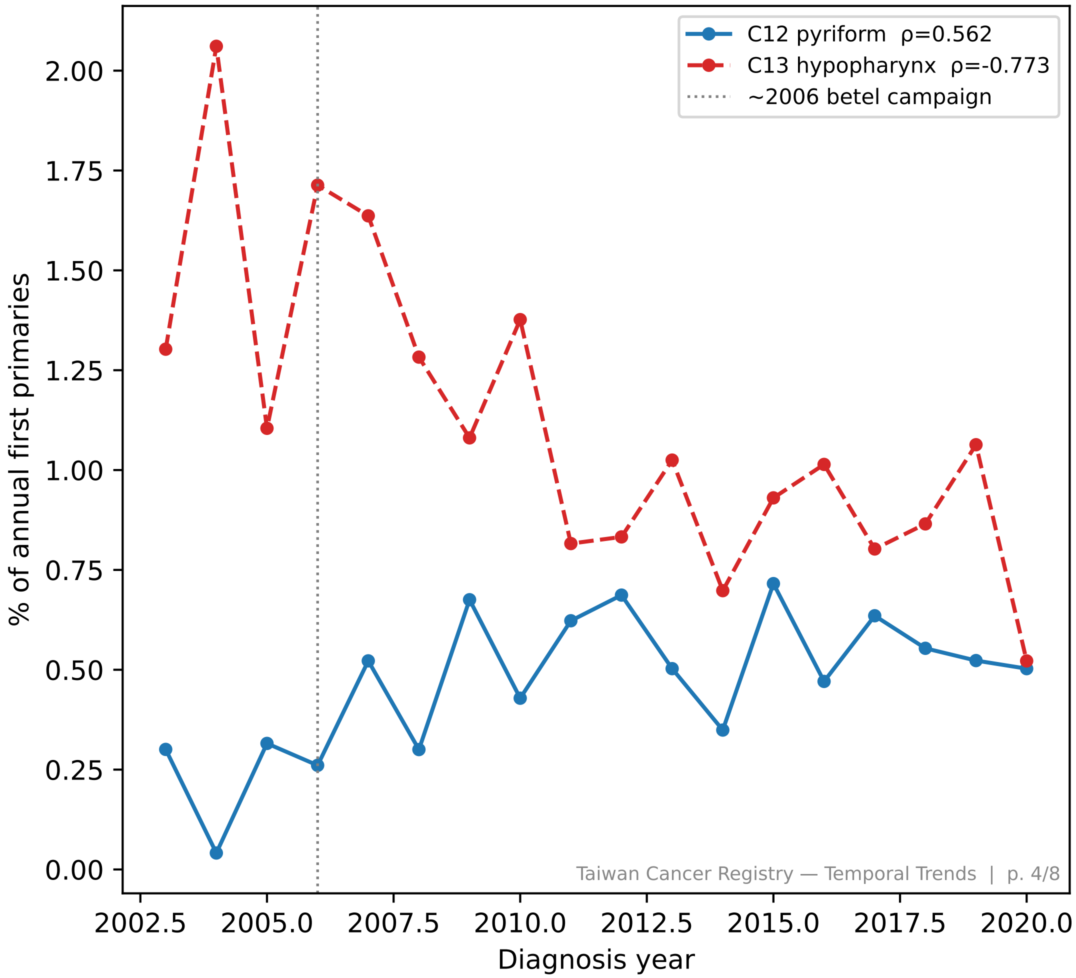


UADT betel-tobacco sites — temporal trends (Hypothesis 2)

UADT sites: annual fraction 2003–2020

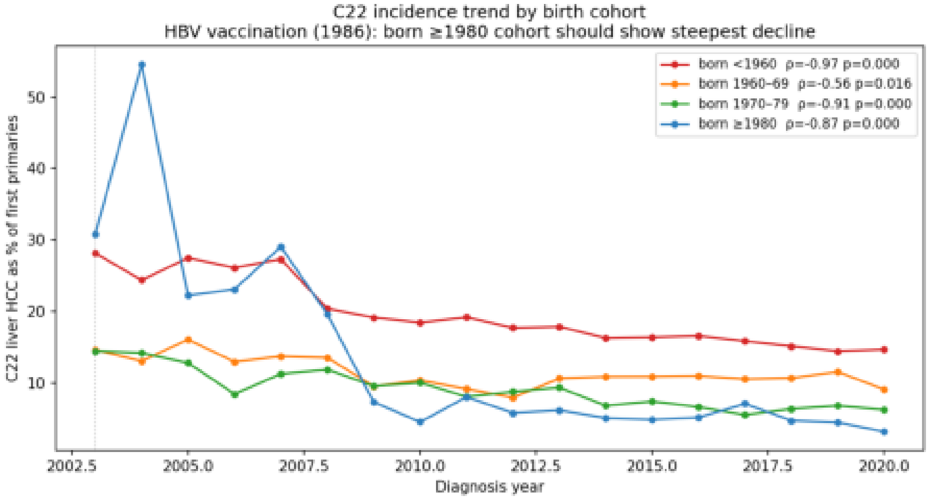


H2 SPLIT: C12 rising vs C13 falling
(pyriform vs hypopharynx — anatomically adjacent, divergent trends)



H1: C22 liver HCC declining — birth-cohort decomposition

C22 liver HCC trend by birth cohort



C22 trend by birth cohort — summary

Birth cohort	N (2003–2020)	ρ	p	Trend
born <1960	50,976	-0.967	0.0000	↓
born 1960–69	16,045	-0.560	0.0156	↓
born 1970–79	8,185	-0.907	0.0000	↓
born ≥1980	3,413	-0.872	0.0000	↓

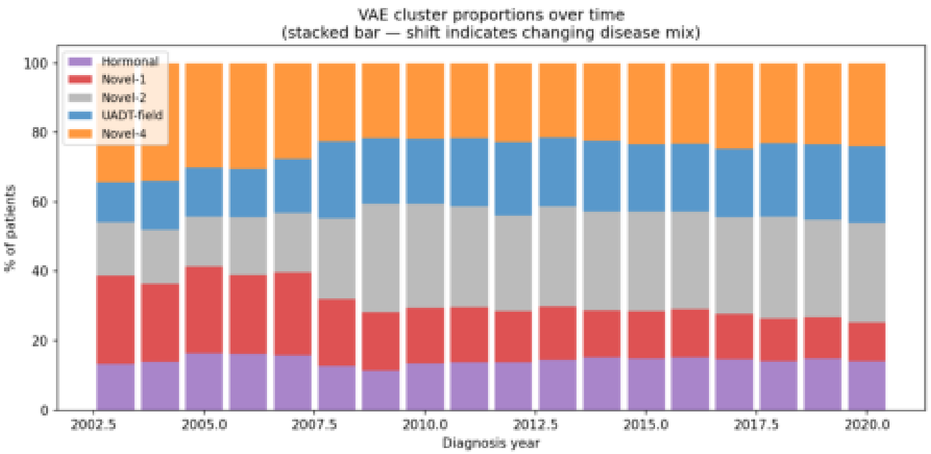
Interpretation

All birth cohorts show declining C22 fraction. The HBV vaccination mechanism predicts the steepest decline in born ≥1980 cohort, but cancer onset at age <40 is rare, limiting statistical power (n=3,413 in this group).

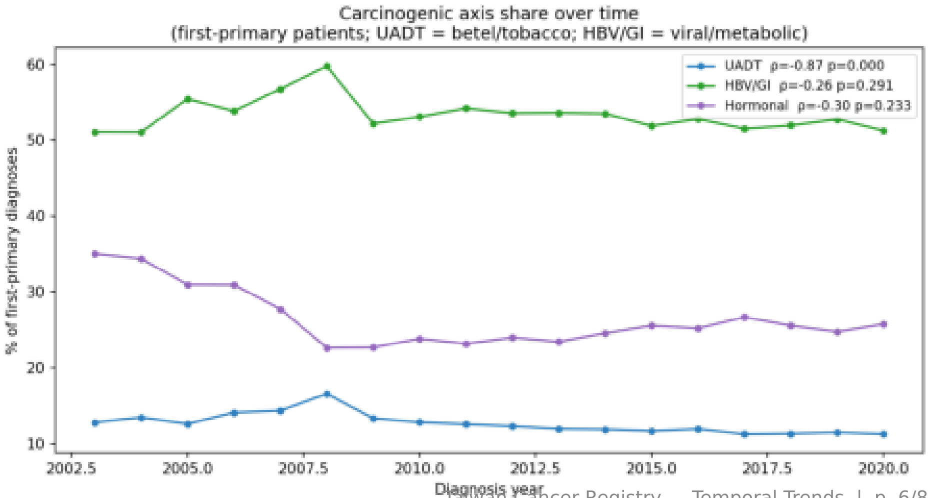
A longer observation window (2030+) will be required to test the vaccination cohort effect conclusively in a cancer registry.

VAE cluster mix and axis share over time

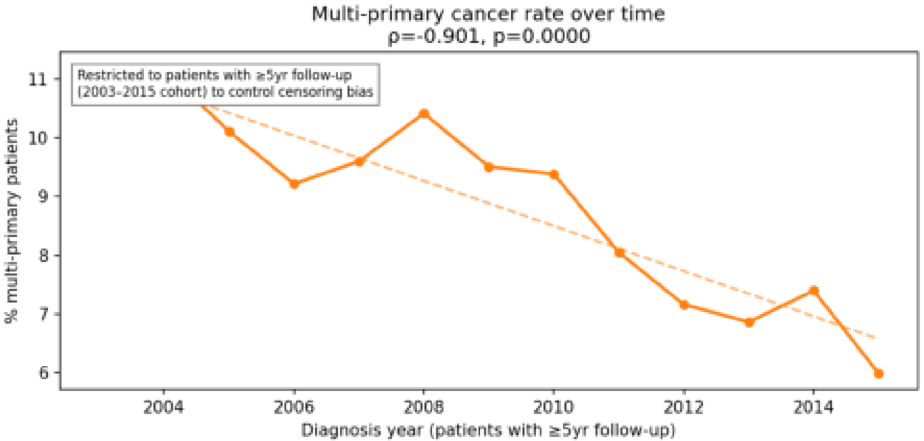
VAE cluster proportion over time



Carcinogenic axis share (first primaries)



Multi-primary rate over time (≥5yr FU cohort)



Top 10 significant trends (5 rising, 5 falling)

Site	Axis	ρ	p_corr	Direction
C54	Hormonal	0.925	0.0085	rising
C61	HBV/GI	0.895	0.0101	rising
C77	Other	0.887	0.0012	rising
C07	Other	0.868	0.0000	rising
C32	Other	0.812	0.0030	rising
C22	HBV/GI	-0.983	0.0001	falling
C06	UADT	-0.909	0.0003	falling
C02	UADT	-0.870	0.0001	falling
C13	UADT	-0.773	0.0000	falling
C05	UADT	-0.692	0.0004	falling

Interpretation and Limitations

Temporal Trends — Taiwan Cancer Registry 2003–2020

Principal Findings

1. HBV vaccination effect (H1 confirmed):
   C22 liver HCC shows $\rho=-0.983$, the single strongest trend in the registry. Novel-1 VAE cluster (which contains C22/C34/C61) mirrors this at $\rho=-0.977$. Both trends are consistent with the cohort effect of Taiwan's 1986 universal infant HBV vaccination programme. Birth-cohort decomposition shows universal decline across age groups; the youngest cohort (born ≥1980) has too few cancer patients in 2003–2020 to test the vaccination mechanism directly.

2. Betel nut regulation (H2 split):
   C13 hypopharynx declining ($\rho=-0.773$) is consistent with Taiwan's anti-betel campaigns from ~2005. C12 pyriform sinus rising ($\rho=+0.562$) is anatomically surprising — possible explanations: differential detection (C12 is often detected incidentally during endoscopy for C15), or anatomical reclassification. Requires clinical review.

3. Metabolic syndrome (H3 partial):
   C54 endometrial rising ($\rho=+0.925$) is the second strongest rising trend and consistent with increasing obesity/metabolic syndrome in Taiwan. C50 breast is flat (ns) — may reflect stable screening coverage or conflicting risk factor trends (later parity, less breastfeeding rising; HRT use declining post-WHI 2002).

4. HPV vaccination / PSA era:
   C53 cervix $\rho=-0.946$ likely reflects gradual benefit of HPV awareness and eventual vaccination; C61 prostate $\rho=+0.895$ likely reflects PSA screening diffusion rather than true incidence increase.

Limitations

- Registry coverage improved 2003–2020 → early years may under-ascertain.
- Betel nut exposure history unavailable — H2 test is ecological.
- Multi-primary rate trend ($\rho=-0.901$, declining) likely reflects residual follow-up bias despite ≥5yr FU restriction; interpret with caution.
- C12 rising vs C13 falling warrants anatomical-pathological sub-study.
- Birth-cohort HBV vaccination test requires ≥2030 registry to reach adequate power in the born ≥1980 cohort.